

GIT/GITHUB BASICS

Lab No	Topic	Content
Lab 1	Git Basics	Setup Git, create repository, add file, commit, pull and push
Lab 2	Git Ignore	Create .gitignore and ignore unwanted files/folders
Lab 3	Branching & Merging	Create branch, modify files, commit and merge with main
Lab 4	Merge Conflict	Create conflict, resolve changes and commit solution
Lab 5	Cleanup & Push	Clean repository, pull latest changes and push to remote

Lab 1

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KIIT0001>git --version
git version 2.46.0.windows.1

C:\Users\KIIT0001> git config --global --list
user.email=suyashchamolil@gmail.com
user.name=Suyash
core.editor=code --wait

C:\Users\KIIT0001>cd desktop

C:\Users\KIIT0001\Desktop>mkdir gitdemo

C:\Users\KIIT0001\Desktop>cd gitdemo

C:\Users\KIIT0001\Desktop\gitdemo>git init
Initialized empty Git repository in C:/Users/KIIT0001/Desktop/gitdemo/.git/

C:\Users\KIIT0001\Desktop\gitdemo>git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

C:\Users\KIIT0001\Desktop\gitdemo>echo welcome to git > welcome.txt

C:\Users\KIIT0001\Desktop\gitdemo>dit
'dit' is not recognized as an internal or external command,
operable program or batch file.
```

```
C:\Users\KIIT0001\Desktop\gitdemo>dir
Volume in drive C is Windows-SSD
Volume Serial Number is EC39-05A9

Directory of C:\Users\KIIT0001\Desktop\gitdemo

20-06-2026  03:30    <DIR>          .
20-06-2026  03:29    <DIR>          ..
20-06-2026  03:30                17 welcome.txt
               1 File(s)                17 bytes
               2 Dir(s) 125,405,192,192 bytes free

C:\Users\KIIT0001\Desktop\gitdemo>type welcome.txt
welcome to git

C:\Users\KIIT0001\Desktop\gitdemo>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    welcome.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\KIIT0001\Desktop\gitdemo>git add .

C:\Users\KIIT0001\Desktop\gitdemo>
C:\Users\KIIT0001\Desktop\gitdemo>git commit -m "created welcome file"
[master (root-commit) cdd5537] created welcome file
 1 file changed, 1 insertion(+)
 create mode 100644 welcome.txt

C:\Users\KIIT0001\Desktop\gitdemo>git status
On branch master
nothing to commit, working tree clean
```

```
C:\Users\KIIT0001\Desktop\gitdemo>git remote add origin https://github.com/Suyashgopal/dummyrepo.git

C:\Users\KIIT0001\Desktop\gitdemo>git remote
origin

C:\Users\KIIT0001\Desktop\gitdemo>git remote -v
origin https://github.com/Suyashgopal/dummyrepo.git (fetch)
origin https://github.com/Suyashgopal/dummyrepo.git (push)

C:\Users\KIIT0001\Desktop\gitdemo>git branch -M main

C:\Users\KIIT0001\Desktop\gitdemo>git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 238 bytes | 119.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Suyashgopal/dummyrepo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\KIIT0001\Desktop\gitdemo>git pull origin main
From https://github.com/Suyashgopal/dummyrepo
 * branch            main       -> FETCH_HEAD
Already up to date.
```

```
C:\Users\KIIT0001\Desktop\gitdemo>git pull origin main
From https://github.com/Suyashgopal/dummyrepo
 * branch            main       -> FETCH_HEAD
Already up to date.

C:\Users\KIIT0001\Desktop\gitdemo>git pull origin main
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 1.86 KiB | 36.00 KiB/s, done.
From https://github.com/Suyashgopal/dummyrepo
 * branch            main       -> FETCH_HEAD
   cdd5537..e2e7d41  main       -> origin/main
Updating cdd5537..e2e7d41
Fast-forward
 1.java      | 1 +
 README.md   | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 1.java
 create mode 100644 README.md

C:\Users\KIIT0001\Desktop\gitdemo>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Lab 2

```
C:\Users\KIIT0001\Desktop\gitdemo>echo error >test.log

C:\Users\KIIT0001\Desktop\gitdemo>mkdir log

C:\Users\KIIT0001\Desktop\gitdemo>ni .gitignore
'ni' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\KIIT0001\Desktop\gitdemo>type nul > .gitignore

C:\Users\KIIT0001\Desktop\gitdemo>echo *.log > .gitignore

C:\Users\KIIT0001\Desktop\gitdemo>echo log/ >> .gitignore

C:\Users\KIIT0001\Desktop\gitdemo>git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .gitignore

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\KIIT0001\Desktop\gitdemo>git add .gitignore

C:\Users\KIIT0001\Desktop\gitdemo>

C:\Users\KIIT0001\Desktop\gitdemo>git commit -m "added git ignore"
[main 80897d9] added git ignore
 1 file changed, 2 insertions(+)
 create mode 100644 .gitignore
```

Lab3

```
C:\Users\KIIT0001\Desktop\gitdemo>git branch feature1

C:\Users\KIIT0001\Desktop\gitdemo>git checkout feature1
Switched to branch 'feature1'

C:\Users\KIIT0001\Desktop\gitdemo>git branches
git: 'branches' is not a git command. See 'git --help'.

C:\Users\KIIT0001\Desktop\gitdemo>git branch
* feature1
  main

C:\Users\KIIT0001\Desktop\gitdemo>git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

C:\Users\KIIT0001\Desktop\gitdemo>git switch feature1
Switched to branch 'feature1'

C:\Users\KIIT0001\Desktop\gitdemo>echo f1 update > feature.txt

C:\Users\KIIT0001\Desktop\gitdemo>git add .

C:\Users\KIIT0001\Desktop\gitdemo>git commit -m"update in branch"
[feature1 4845c19] update in branch
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

C:\Users\KIIT0001\Desktop\gitdemo>git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
```

```
C:\Users\KIIT0001\Desktop\gitdemo>git diff feature1
diff --git a/feature.txt b/feature.txt
deleted file mode 100644
index 6f6c2f1..0000000
--- a/feature.txt
+++ /dev/null
@@ -1,0 @@
-f1 update

C:\Users\KIIT0001\Desktop\gitdemo>git merge feature1
Updating 80897d9..4845c19
Fast-forward
 feature.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

C:\Users\KIIT0001\Desktop\gitdemo>git log --oneline --graph --update
fatal: unrecognized argument: --update

C:\Users\KIIT0001\Desktop\gitdemo>git log --oneline --graph --decorate
* 4845c19 (HEAD -> main, feature1) update in branch
* 80897d9 added git ignore
* e2e7d41 (origin/main) Create 1.java
* a6af6dd Update README with test information
* cdd5537 created welcome file

C:\Users\KIIT0001\Desktop\gitdemo>git branch -d feature1
Deleted branch feature1 (was 4845c19).
```

Lab4

```
Command Prompt

C:\Users\KIIT0001\Desktop\gitdemo>git checkout -b newbranch
Switched to a new branch 'newbranch'

C:\Users\KIIT0001\Desktop\gitdemo>echo branch text > b1.txt

C:\Users\KIIT0001\Desktop\gitdemo>git add .

C:\Users\KIIT0001\Desktop\gitdemo>git commit -m "branch update for conflict"
[newbranch 3f634dc] branch update for conflict
1 file changed, 1 insertion(+)
create mode 100644 b1.txt

C:\Users\KIIT0001\Desktop\gitdemo>git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
(use "git push" to publish your local commits)

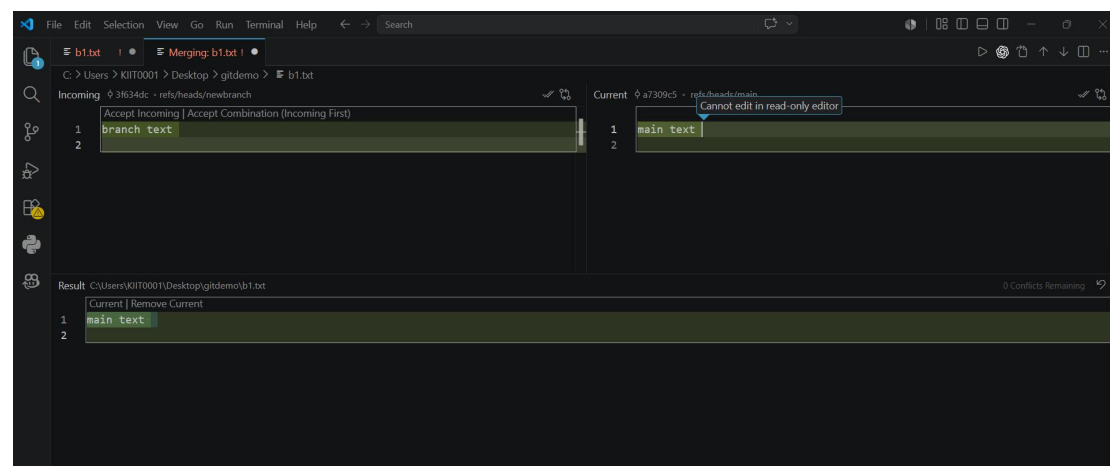
C:\Users\KIIT0001\Desktop\gitdemo>echo main text > b1.txt

C:\Users\KIIT0001\Desktop\gitdemo>git add .

C:\Users\KIIT0001\Desktop\gitdemo>git commit -m "main branch made update to the same file present in newbranch"
[main a7309c5] main branch made update to the same file present in newbranch
1 file changed, 1 insertion(+)
create mode 100644 b1.txt

C:\Users\KIIT0001\Desktop\gitdemo>git merge newbranch
Auto-merging b1.txt
CONFLICT (add/add): Merge conflict in b1.txt
Automatic merge failed; fix conflicts and then commit the result.

C:\Users\KIIT0001\Desktop\gitdemo>code b1.txt
```




```

C:\Users\KIIT0001\Desktop\gitdemo>git commit -m "resoved conflict in vs code"
U
  b1.txt
error: Committing is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.

C:\Users\KIIT0001\Desktop\gitdemo>git merge newbranch
error: Merging is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.

C:\Users\KIIT0001\Desktop\gitdemo>git status
On branch main
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both added:   b1.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\KIIT0001\Desktop\gitdemo>git add b1.txt

C:\Users\KIIT0001\Desktop\gitdemo>git commit -m "resoved conflict in vs code"
[main c8a8bc4] resoved conflict in vs code

```

Lab 5

```

C:\Users\KIIT0001\Desktop\gitdemo>git pull origin main
From https://github.com/Suyashgopal/dummyrepo
 * branch            main       -> FETCH_HEAD
Already up to date.

C:\Users\KIIT0001\Desktop\gitdemo>pit push origin main
'pit' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\KIIT0001\Desktop\gitdemo>git push origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (15/15), 1.27 KiB | 325.00 KiB/s, done.
Total 15 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To https://github.com/Suyashgopal/dummyrepo.git
 e2e7d41..c8a8bc4  main -> main

```